Task 2 — Supervised Baseline Results

Section 1 — Top-1 Supervised Result (SSL Reference Baseline)

Rank	Model	Best Split	Accuracy	Macro F1	GFLOPs	Params	Train Time	Test Time
1	**MobileNet-V3-Large**	90:10	**92.16%**	74.75%	0.467	4.2M	29.2 min	2.18 s
2	ViT-B/16	90:10	91.18%	74.87%	33.695	85.6M	87.1 min	3.28 s
3	ResNet-101	80:20	87.25%	71.32%	15.729	42.5M	41.8 min	4.16 s
4	ResNet-50	90:10	85.29%	69.81%	8.263	23.5M	33.5 min	3.59 s
5	EfficientNet-B0	90:10	84.31%	69.81%	0.828	4.0M	31.5 min	2.14 s

Section 2 — Accuracy Grid (All 45 Experiments)

Model	90:10	80:20	70:30	60:40	50:50	40:60	30:70	20:80	10:90
EfficientNet-B0	**84.3%**	79.9%	78.4%	75.7%	76.0%	73.3%	71.3%	60.8%	49.2%
MobileNet-V3-Large	**92.2%**	79.9%	79.0%	78.9%	69.1%	73.9%	63.8%	42.4%	17.4%
ResNet-50	**85.3%**	76.5%	84.3%	76.7%	78.7%	72.6%	71.9%	56.1%	33.4%
ResNet-101	87.3%	**87.3%**	79.0%	79.4%	79.3%	69.2%	68.8%	62.0%	37.7%
ViT-B/16	**91.2%**	89.7%	84.9%	87.2%	82.3%	81.5%	82.4%	68.8%	49.0%

Section 3 — Per-Model Detailed Results

Model 1: MobileNet-V3-Large

Architecture	Pretrained	Parameters	GFLOPs @ 224×224
Inverted residuals + Squeeze-Excitation blocks	ImageNet	4.2M	0.467

Best Performance: 92.16% accuracy @ 90:10 split | Macro F1: 74.75%

Key Strength: Highest accuracy with lowest GFLOPs (0.467) — optimal efficiency. Designated SSL reference baseline.

Top Confused Pairs: Pratylenchus→Hirschmaniella | Hemicycliophora→Pratylenchus

Split	Train N	Test N	Accuracy	Macro F1	ROC-AUC	Train Time	Test Time
90:10	822	102	**92.16%**	74.75%	N/A	29.2 min	2.18 s
80:20	731	203	79.90%	67.69%	N/A	9.3 min	3.84 s
70:30	640	305	79.02%	63.91%	0.9435	8.8 min	5.18 s
60:40	549	407	78.87%	68.55%	0.9101	7.4 min	6.55 s
50:50	457	508	69.09%	57.72%	0.9503	6.0 min	7.90 s
40:60	366	610	73.93%	69.35%	0.9345	5.4 min	10.70 s
30:70	275	711	63.76%	58.94%	0.9148	4.0 min	12.61 s
20:80	183	813	42.44%	34.09%	0.7711	3.2 min	13.96 s
10:90	92	914	17.38%	9.28%	0.6322	1.7 min	21.89 s

Performance Scaling with Training Data

Training %	Accuracy	Macro F1	Change
10%	17.38%	9.28%	—
20%	42.44%	34.09%	+25.06%
30%	63.76%	58.94%	+21.32%
40%	73.93%	69.35%	+10.17%
50%	69.09%	57.72%	−4.84% (dip)
60%	78.87%	68.55%	+9.78%
70%	79.02%	63.91%	+0.15%
80%	79.90%	67.69%	+0.88%
90%	**92.16%**	**74.75%**	+12.26%

Model 2: ViT-B/16

Architecture	Pretrained	Parameters	GFLOPs @ 224×224
Vision Transformer (patch size 16)	ImageNet-21K	85.6M	33.695

Best Performance: 91.18% accuracy @ 90:10 split | Macro F1: 74.87%

Key Strength: Most consistent across middle splits (30:70→70:30); highest Macro F1 (74.87%)

Top Confused Pairs: Helicotylenchus→Meloidogyne | Hirschmaniella→Meloidogyne

Split	Train N	Test N	Accuracy	Macro F1	ROC-AUC	Train Time	Test Time
90:10	822	102	**91.18%**	74.87%	N/A	87.1 min	3.28 s
80:20	731	203	89.71%	74.16%	N/A	81.9 min	4.72 s
70:30	640	305	84.92%	70.00%	0.9089	60.1 min	6.51 s
60:40	549	407	87.22%	70.67%	0.9081	56.2 min	8.40 s
50:50	457	508	82.28%	69.16%	0.9684	44.1 min	11.07 s
40:60	366	610	81.48%	74.99%	0.9480	37.6 min	12.75 s
30:70	275	711	82.44%	72.01%	0.9783	34.2 min	14.08 s
20:80	183	813	68.76%	56.72%	0.9095	21.0 min	12.51 s
10:90	92	914	48.96%	39.85%	0.8098	10.9 min	14.68 s

Performance Scaling with Training Data

Training %	Accuracy	Macro F1
10%	48.96%	39.85%
20%	68.76%	56.72%
30%	82.44%	72.01%
40%	81.48%	74.99%
50%	82.28%	69.16%
60%	87.22%	70.67%
70%	84.92%	70.00%
80%	89.71%	74.16%
90%	**91.18%**	**74.87%**

Model 3: ResNet-101

Architecture	Pretrained	Parameters	GFLOPs @ 224×224
ResNet-101 deep residual blocks	ImageNet	42.5M	15.729

Best Performance: 87.25% accuracy @ 80:20 split | Macro F1: 71.32%
Key Strength: Most consistent in 50–90% training range; best F1 at 50:50 (71.78%)
Top Confused Pairs: Pratylenchus→Hirschmaniella | Xiphinema→Hirschmaniella

Split	Train N	Test N	Accuracy	Macro F1	ROC-AUC	Train Time	Test Time
90:10	822	102	87.25%	73.37%	N/A	44.0 min	2.56 s
80:20	731	203	**87.25%**	71.32%	N/A	41.8 min	4.16 s
70:30	640	305	79.02%	64.41%	0.9114	32.3 min	7.37 s
60:40	549	407	79.36%	64.00%	0.8940	29.1 min	7.88 s
50:50	457	508	79.33%	71.78%	0.9667	27.1 min	11.68 s
40:60	366	610	69.18%	57.36%	0.9220	22.2 min	11.95 s
30:70	275	711	68.82%	55.62%	0.8930	16.0 min	23.03 s
20:80	183	813	61.99%	52.57%	0.8669	11.9 min	15.67 s
10:90	92	914	37.70%	27.72%	0.7432	7.7 min	16.08 s

Model 4: ResNet-50

Architecture	Pretrained	Parameters	GFLOPs @ 224×224
ResNet-50 residual blocks	ImageNet	23.5M	8.263

Best Performance: 85.29% accuracy @ 90:10 split | Macro F1: 69.81%
Key Strength: Consistent ROC-AUC across splits (0.716–0.963)
Top Confused Pairs: Meloidogyne→Hirschmaniella | Helicotylenchus→Pratylenchus

Split	Train N	Test N	Accuracy	Macro F1	ROC-AUC	Train Time	Test Time
90:10	822	102	**85.29%**	69.81%	N/A	33.5 min	3.59 s
80:20	731	203	76.47%	64.33%	N/A	31.2 min	4.76 s
70:30	640	305	84.26%	69.36%	0.9060	28.3 min	6.94 s
60:40	549	407	76.66%	62.48%	0.9296	22.9 min	6.84 s
50:50	457	508	78.74%	64.37%	0.9625	19.9 min	11.19 s
40:60	366	610	72.62%	67.51%	0.9295	18.6 min	11.44 s
30:70	275	711	71.91%	65.16%	0.9306	14.3 min	11.86 s
20:80	183	813	56.09%	48.19%	0.8885	11.3 min	13.96 s
10:90	92	914	33.44%	25.66%	0.7163	5.9 min	13.12 s

Model 5: EfficientNet-B0

Architecture	Pretrained	Parameters	GFLOPs @ 224×224
EfficientNet compound scaling	ImageNet	4.0M	0.828

Best Performance: 84.31% accuracy @ 90:10 split | Macro F1: 69.81%
Key Strength: Lowest GFLOPs (0.828) — best efficiency-accuracy trade-off
Top Confused Pairs: Hirschmaniella→Meloidogyne | Pratylenchus→Criconemoides

Split	Train N	Test N	Accuracy	Macro F1	ROC-AUC	Train Time	Test Time
90:10	822	102	**84.31%**	69.81%	N/A	31.5 min	2.14 s
80:20	731	203	79.90%	66.23%	N/A	10.8 min	3.91 s
70:30	640	305	78.36%	63.75%	0.8504	8.9 min	3.76 s
60:40	549	407	75.68%	62.94%	0.8935	7.4 min	7.00 s
50:50	457	508	75.98%	62.48%	0.9412	6.0 min	8.70 s
40:60	366	610	73.28%	68.75%	0.9373	5.3 min	10.07 s
30:70	275	711	71.35%	63.33%	0.9167	4.0 min	10.18 s
20:80	183	813	60.76%	53.83%	0.8520	2.9 min	14.17 s
10:90	92	914	49.18%	40.20%	0.7598	39.0 min	35.77 s

Section 4 — Comparative Analysis

4.1 Overall Performance Ranking

Model	Accuracy	Macro F1	GFLOPs	Params	Train Time	Test Time	Best For
MobileNet-V3-L	**92.16%**	74.75%	0.467	4.2M	29.2 min	2.18 s	SSL Baseline / Deployment
ViT-B/16	91.18%	74.87%	33.695	85.6M	87.1 min	3.28 s	Best generalisation
ResNet-101	87.25%	71.32%	15.729	42.5M	41.8 min	4.16 s	Mid-split consistency
ResNet-50	85.29%	69.81%	8.263	23.5M	33.5 min	3.59 s	Balanced baseline
EfficientNet-B0	84.31%	69.81%	0.828	4.0M	31.5 min	2.14 s	Edge / low-compute

4.2 Memory & Compute

Model	Parameters	Memory	GFLOPs
EfficientNet-B0	4.0M	~15 MB	0.828
MobileNet-V3-Large	4.2M	~16 MB	0.467
ResNet-50	23.5M	~90 MB	8.263
ResNet-101	42.5M	~162 MB	15.729
ViT-B/16	85.6M	~327 MB	33.695

4.3 Training Time at 90:10 Split

EfficientNet-B0 : 31.5 min [redacted]
MobileNet-V3-L : 29.2 min [redacted]
ResNet-50 : 33.5 min [redacted]
ResNet-101 : 44.0 min [redacted]
ViT-B/16 : 87.1 min [redacted]

Section 5 — Failure Mode Analysis

All models share consistent failure patterns — morphologically similar genera are most often confused across all splits.

Model	#1 Confused Pair	#2 Confused Pair	Pattern
EfficientNet-B0	Hirschmaniella→Meloidogyne	Pratylenchus→Criconemoides	Shape similarity
MobileNet-V3-Large	Pratylenchus→Hirschmaniella	Hemicycliophora→Pratylenchus	Vermiform confusion
ResNet-50	Meloidogyne→Hirschmaniella	Helicotylenchus→Pratylenchus	Spiral vs. vermiform
ResNet-101	Pratylenchus→Hirschmaniella	Xiphinema→Hirschmaniella	Stylet confusion
ViT-B/16	Helicotylenchus→Meloidogyne	Hirschmaniella→Meloidogyne	Global shape confusion

Section 6 — SSL Baseline Declaration

Metric	Value
Chosen baseline	MobileNet-V3-Large @ 90:10 split
Supervised Accuracy	92.16%
Macro F1	74.75%
GFLOPs	0.467 (most efficient top performer)
Parameters	4.2M

Section 7 — Experimental Setup

Setting	Value
Dataset	Microscopic Image Dataset of Plant-Parasitic Nematodes
Dataset DOI	https://data.mendeley.com/datasets/cck8yxj3xw (v2)
Total images	1,016 images across 11 genera
Input size	224 × 224 pixels
Splits tested	90:10, 80:20, 70:30, 60:40, 50:50, 40:60, 30:70, 20:80, 10:90
Validation	10% of training portion (group-aware GroupShuffleSplit)
Optimizer	AdamW (lr = 3×10^{-4})
Loss function	Cross-Entropy with class weights
Min epochs	50 (early stopping, patience = 8)
Batch size	32
Pretrained weights	ImageNet (torchvision DEFAULT)
Platform	Kaggle GPU — NVIDIA T4 / P100
Random seed	42 (fixed for reproducibility)